INCH-POUND

MIL-M-38510/370A
15 July 2004
SUPERSEDING
MIL-M-38510/370(USAF)
18 March 1983

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED LOW POWER SCHOTTKY TTL,
NAND GATES, MONOLITHIC SILICON

Inactive for new design after 8 July 1997.

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF 38535

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, advanced low power Schottky TTL, positive NAND logic gating microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number. For this product, the requirements of MIL-M-38510 have been superseded by MIL-PRF-38535, (see 6.3).

1.2 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-38535, and as specified herein.

1.2.1 Device types. The device types are as follows:

Device type	Circuit
01	Quadruple, 2-input positive NAND gate
02	Triple, 3-input positive NAND gate
03	Dual, 4-input positive NAND gate
04	Single, 8-input positive NAND gate
05	Single, 13-input positive NAND gate
06	Hex, 1-input inverter gate

1.2.2 Device class. The device class is the product assurance level as defined in MIL-PRF-38535.

1.2.3 Case outlines. The case outlines are as designated in MIL-STD-1835 and as follows:

Outline letter	Descriptive designator	Terminals	Package style
A	GDFP5-F14 or CDFP6-F14	14	Flat pack
B	GDFP4-14	14	Flat pack
C	GDIP1-T14 or CDIP2-T14	14	Dual-in-line
D	GDFP1-F14 or CDFP2-F14	14	Flat pack
E	GDIP1-T16 or CDIP2-T16	16	Dual-in-line
F	GDFP2-F16 or CDFP3-F16	16	Flat pack
2	CQCC1-N20	20	Square leadless chip carrier

Comments, suggestions, or questions on this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAS, P. O. Box 3990, Columbus, OH 43218-3990, or emailed to bipolar@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

AMSC N/A FSC 5962

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to +7.0 V dc
Storage temperature range	-65° to +150°C
Maximum power dissipation (P_D), per device: 1/	
Device 01	16.4 mW
Device 02	12.3 mW
Device 03	8.2 mW
Device 04 and 05	4.1 mW
Device 06	24.6 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction to case (θ_{JC}):	
Cases A, B, C, D, E, F, and 2	(See MIL-STD-1835)
Junction temperature (T_J) 2/	175°C

1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH})	2.0 V dc
Maximum low level input voltage (V_{IL})	0.8 V dc
Case operating temperature range (T_C)	-55° to +125°C

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-38535 - Integrated Circuits (Microcircuits) Manufacturing, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-883 - Test Method Standard for Microelectronics.
MIL-STD-1835 - Interface Standard Electronic Component Case Outlines

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

1/ Must withstand the added P_D due to short-circuit test (e.g., I_O).
2/ Maximum junction temperature should not be exceeded except in accordance with allowable short duration burn-in screening condition in accordance with MIL-PRF-38535.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Qualification. Microcircuits furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.4).

3.2 Item requirements. The individual item requirements shall be in accordance with MIL-PRF-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-38535 and herein.

3.3.1 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3.2 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1.

3.3.3 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.3.4 Schematic circuits. The schematic circuits shall be maintained by the manufacturer and made available to the qualifying activity and the preparing activity upon request.

3.4 Lead material and finish. The lead material and finish shall be in accordance with MIL-PRF-38535 (see 6.6).

3.5 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.7 Marking. Marking shall be in accordance with MIL-PRF-38535.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 8 (see MIL-PRF-38535, appendix A).

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein.

4.2 Screening. Screening shall be in accordance with MIL-PRF-38535 and shall be conducted on all devices prior to qualification and conformance inspection. The following additional criteria shall apply:

a. The burn-in test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1015 of MIL-STD-883.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. Additional screening for space level product shall be as specified in MIL-PRF-38535.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	Limits		Unit
				Min	Max	
High level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IL} = 0.8 V, I_{OH} = -400 µA	All	2.5		V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 4 mA, V_{IH} = 2.0 V	All		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA, T_C = +25°C	All		-1.5	V
High level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IH} = 2.7 V	All		20	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IH} = 7.0 V	All		100	µA
Low level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	All	0	-400	µA
Output current 1/	I_O	V_{CC} = 5.5 V, V_O = 2.25 V	01, 03, 06	-15	-110	mA
			02, 04, 05	-20	-112	
High level supply current	I_{CCH}	V_{CC} = 5.5 V, V_{IN} = 0 V	01		0.85	mA
			02		0.6	
			03		0.4	
			04		0.36	
			05		0.34	
			06		1.1	
Low level supply current	I_{CCL}	V_{CC} = 5.5 V, V_{IN} = 4.5 V	01		3.0	mA
			02		2.2	
			03		1.5	
			04		0.9	
			05		0.8	
			06		4.2	

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	Limits		Unit
				Min	Max	
Propagation delay time, high-to-low level	t_{PHL}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 500 Ω	01	2	10	ns
			02	3	12	
			03	3	15	
			04	5	22	
			05	5	28	
			06	2	12	
Propagation delay time low-to-high level	t_{PLH}		01	3	14	ns
			02	3	15	
			03	3	13	
			04	3	12	
			05	3	14	
			06	3	17	

1/ The output conditions have been chosen to produce a current that closely approximates one half of the true short-circuit output current, I_{OS}.

TABLE II. Electrical test requirements.

MIL-PRF-38535 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters	1	1
Final electrical test parameters	1*, 2, 3, 9, 10, 11	1*, 2, 3, 9
Group A test requirements	1, 2, 3, 9, 10, 11	1, 2, 3, 9, 10, 11
Group B electrical test parameters when using the method 5005 QCI option	1, 2, 3, 9, 10, 11	N/A
Group C end-point electrical parameters	1, 2, 3, 9, 10, 11	1, 2, 3
Group D end-point electrical parameters	1, 2, 3	1, 2, 3

*PDA applies to subgroup 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-38535.

4.4 Technology Conformance Inspection (TCI). Technology conformance inspection shall be in accordance with MIL-PRF-38535 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table III of MIL-PRF-38535 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, 6, 7, and 8 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of MIL-PRF-38535.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table IV of MIL-PRF-38535 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. The steady-state life test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table V of MIL-PRF-38535. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

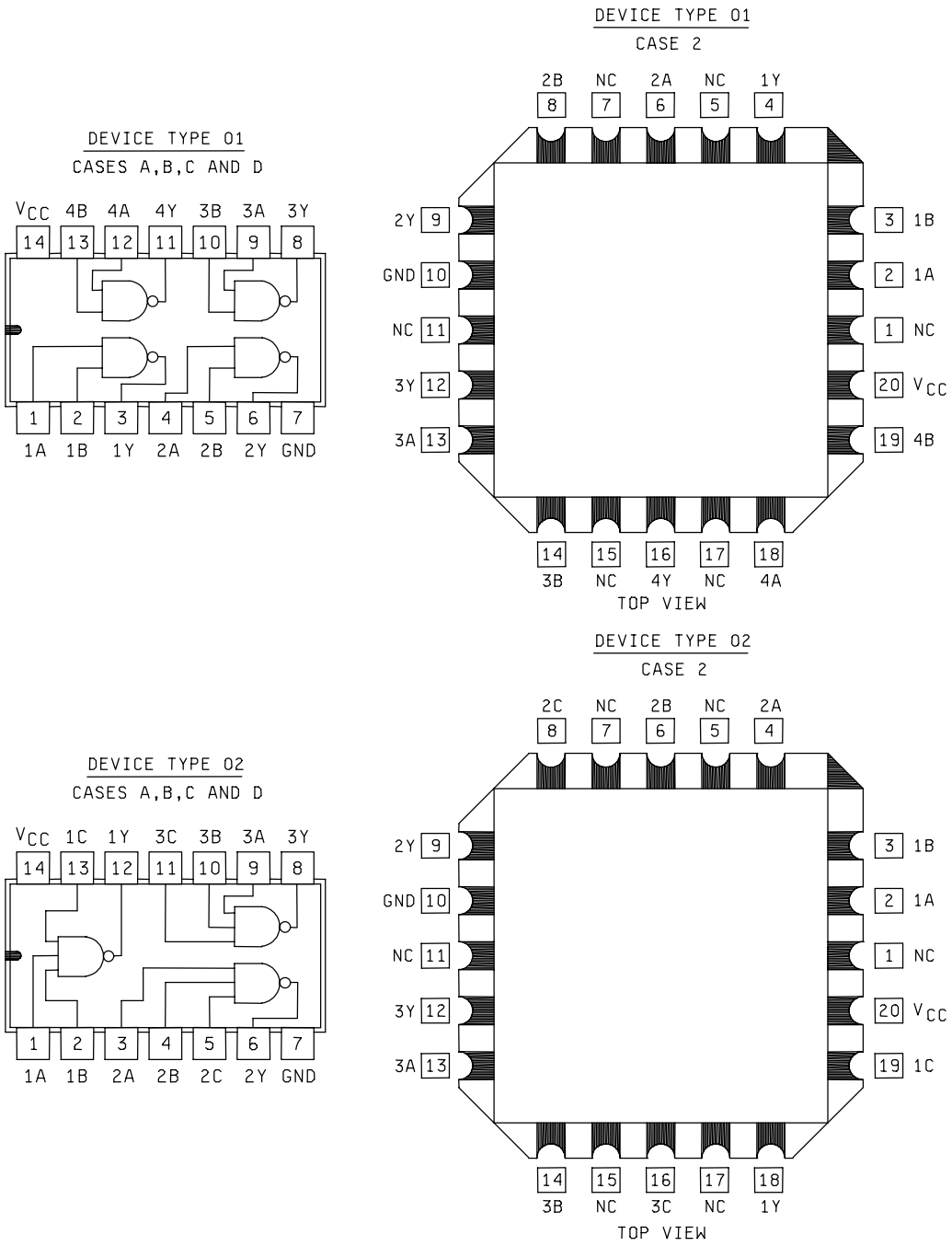


FIGURE 1. Logic diagram and terminal connections (top views).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

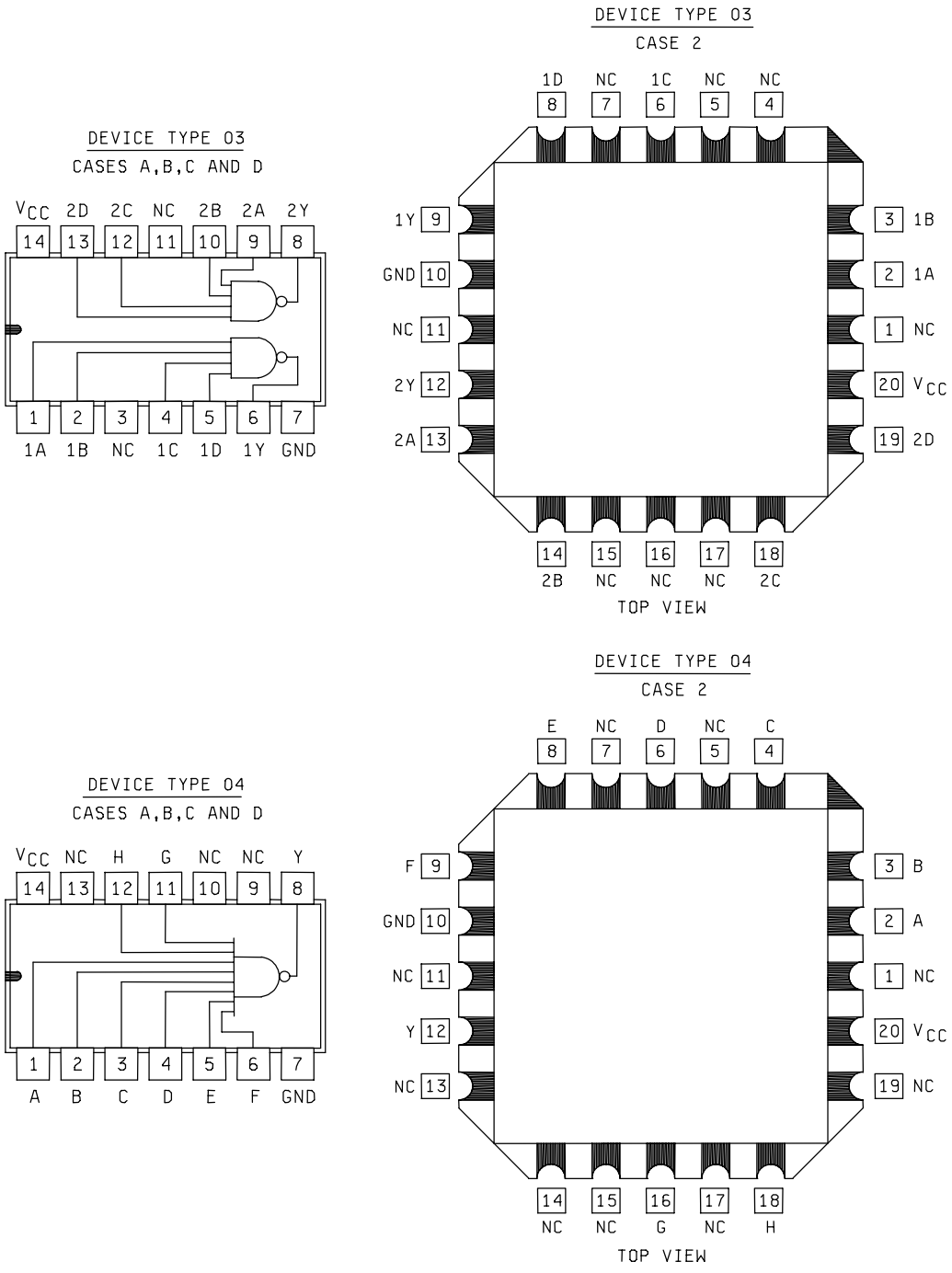


FIGURE 1. Logic diagram and terminal connections (top views) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

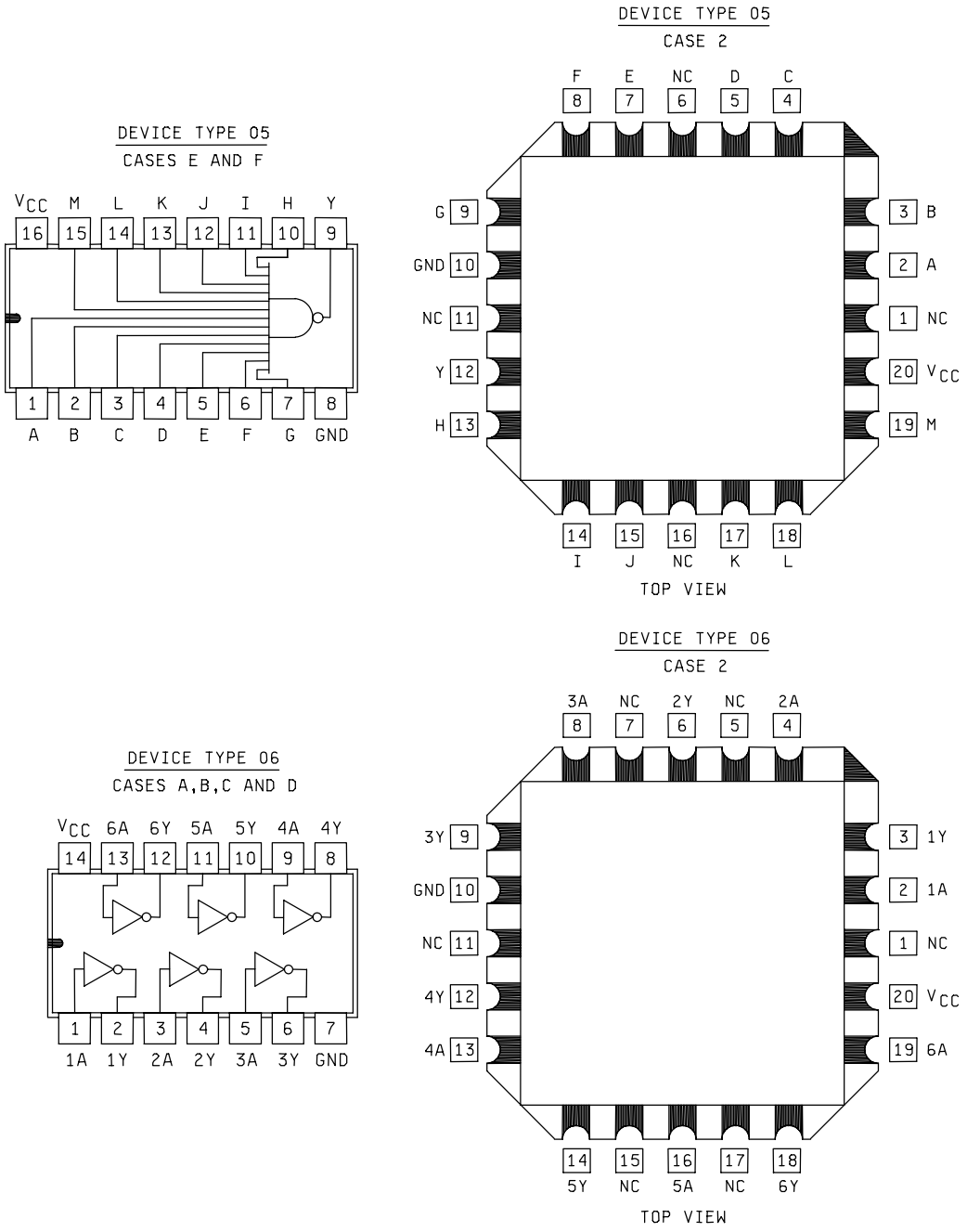


FIGURE 1. Logic diagram and terminal connections (top views) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

Device type 01

Truth table each gate		
Input		Output
A	B	Y
L	L	H
H	L	H
L	H	H
H	H	L

Positive logic Y = $\overline{AB}$

Device type 02

Truth table each gate			
Input			Output
A	B	C	Y
L	L	L	H
H	L	L	H
L	H	L	H
H	H	L	H
L	L	H	H
H	L	H	H
L	H	H	H
H	H	H	L

Positive logic Y = $\overline{ABC}$

Device type 03

Input				Output
A	B	C	D	Y
L	L	L	L	H
H	L	L	L	H
L	H	L	L	H
H	H	L	L	H
L	L	H	L	H
H	L	H	L	H
L	H	H	L	H
H	H	H	L	H
L	L	L	H	H
H	L	L	H	H
L	H	L	H	H
H	H	L	H	H
L	L	H	H	H
H	L	H	H	H
L	H	H	H	H
H	H	H	H	L

Positive logic Y = $\overline{ABCD}$

Device type 04

Truth table								
Inputs								Outputs
A	B	C	D	E	F	G	H	Y
H	H	H	H	H	H	H	H	L
All other combinations of H and L at the inputs give H output.								

Positive logic Y = $\overline{ABCDEFGH}$

FIGURE 2. Truth tables and logic equations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

Device type 05

Truth table													
Inputs													Outputs
A	B	C	D	E	F	G	H	I	J	K	L	M	Y
H	H	H	H	H	H	H	H	H	H	H	H	H	L
All other combinations of H and L at the inputs give H output.													

Positive logic $Y = \overline{ABCDEFGHIJKLM}$

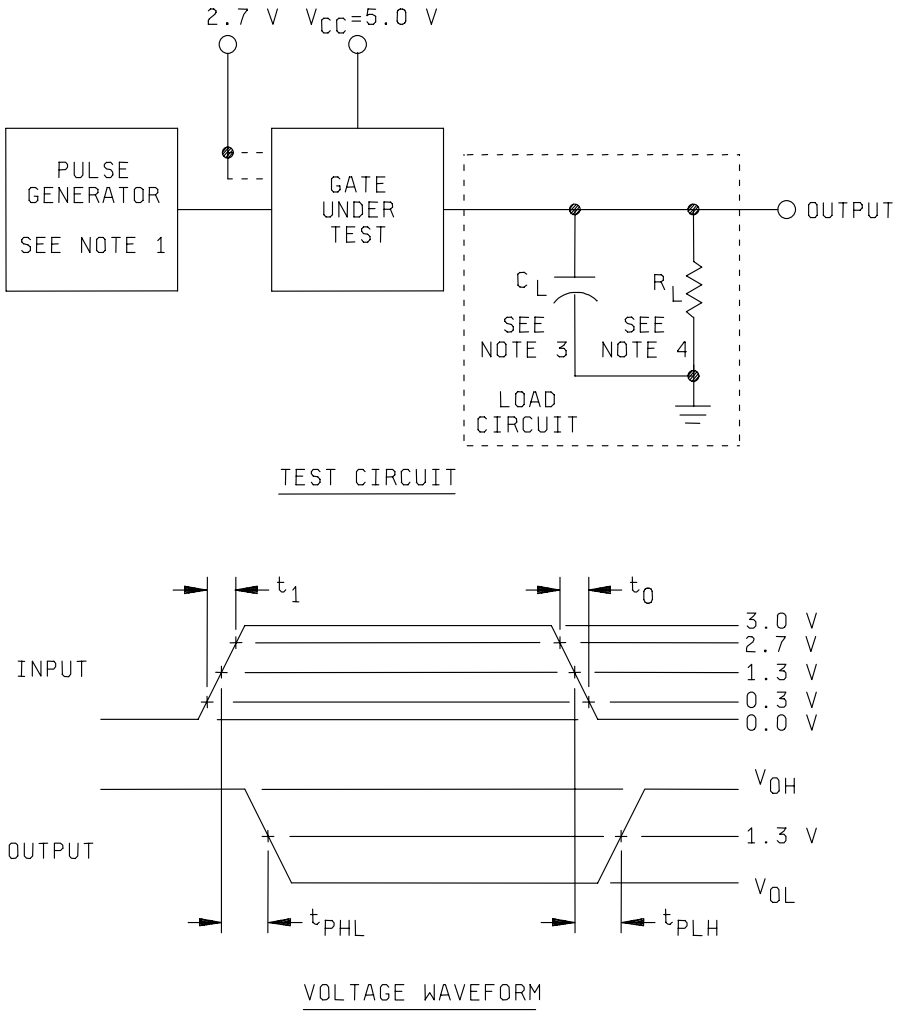
Device type 06

Truth table each gate	
A	Y
L	H
H	L

Positive logic $Y = \overline{A}$

FIGURE 2. Truth tables and logic equations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.



NOTES:

1. Input pulse has the following characteristics:
   $t_1 = t_0 = 6 \pm 1.5$ ns, PRR $\leq$ 1.0 MHz, $Z_{OUT} \cong 50\Omega$.
2. Inputs not under test are at 2.7 V.
3. C_L = 50 pF ±10%, including scope probe, wiring and stray capacitance without package in test fixture.
4. $R_L = 499\Omega \pm 1\%$.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 3. Switching time test circuit and waveforms for device types 01 through 06.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	1A	1B	1Y	2A	2B	2Y	GND	3Y	3A	3B	4Y	4A	4B	V_{CC}		Min	Max	
1 Tc = 25°C	V_{OH}	3006	1	5.5 V	0.8 V	-400 µA	5.5 V	5.5 V		GND		5.5 V	5.5 V		5.5 V	5.5 V	4.5 V	1Y	2.5		V
		"	2	0.8 V	5.5 V	-400 µA	"	5.5 V		"		"	"		"	"	"	1Y	"		"
		"	3	5.5 V	"		"	0.8 V	-400 µA	"		"	"		"	"	"	2Y	"		"
		"	4	"	"		0.8 V	5.5 V	-400 µA	"		"	"		"	"	"	2Y	"		"
		"	5	"	"		5.5 V	"		"	-400 µA	"	0.8 V		"	"	"	3Y	"		"
		"	6	"	"		"	"		"	-400 µA	0.8 V	5.5 V		"	"	"	3Y	"		"
		"	7	"	"		"	"		"		5.5 V	"	-400 µA	"	0.8 V	"	4Y	"		"
		"	8	"	"		"	"		"		5.5 V	"	-400 µA	0.8 V	5.5 V	"	4Y	"		"
	V_{OL}	3007	9	2.0 V	2.0 V	4 mA	GND	GND		"		GND	GND		GND	GND	"	1Y		0.4	"
		"	10	GND	GND		2.0 V	2.0 V	4 mA	"		GND	GND		"	"	"	2Y		"	"
		"	11	"	"		GND	GND		"	4 mA	2.0 V	2.0 V		"	"	"	3Y		"	"
		"	12	"	"		GND	GND		"		GND	GND	4 mA	2.0 V	2.0 V	"	4Y		"	"
	$V_{I C}$		13	-18 mA						"							"	1A		-1.5	"
			14		-18 mA					"							"	1B		"	"
			15				-18 mA			"							"	2A		"	"
			16					-18 mA		"							"	2B		"	"
			17							"		-18 mA					"	3A		"	"
			18							"			-18 mA				"	3B		"	"
			19							"					-18 mA		"	4A		"	"
			20							"						-18 mA	"	4B		"	"
	I_{IH1}	3010	21	2.7 V	GND		GND	GND		"		GND	GND		GND	GND	5.5 V	1A		20	µA
		"	22	GND	2.7 V		GND	"		"		"	"		"	"	"	1B		"	"
		"	23	"	GND		2.7 V	"		"		"	"		"	"	"	2A		"	"
		"	24	"	"		GND	2.7 V		"		"	"		"	"	"	2B		"	"
		"	25	"	"		"	GND		"		2.7 V	"		"	"	"	3A		"	"
		"	26	"	"		"	"		"		GND	2.7 V		"	"	"	3B		"	"
		"	27	"	"		"	"		"		"	GND		2.7 V	"	"	4A		"	"
		"	28	"	"		"	"		"		"	"		GND	2.7 V	"	4B		"	"
	I_{IH2}	"	29	7.0 V	"		"	"		"		"	"		"	GND	"	1A		100	"
		"	30	GND	7.0 V		"	"		"		"	"		"	"	"	1B		"	"
		"	31	"	GND		7.0 V	"		"		"	"		"	"	"	2A		"	"
		"	32	"	"		GND	7.0 V		"		"	"		"	"	"	2B		"	"
		"	33	"	"		"	GND		"		7.0 V	"		"	"	"	3A		"	"
		"	34	"	"		"	"		"		GND	7.0 V		"	"	"	3B		"	"
		"	35	"	"		"	"		"		"	GND		7.0 V	"	"	4A		"	"
		"	36	"	"		"	"		"		"	"		GND	7.0 V	"	4B		"	"
	I_{IL}	3009	37	0.4 V	5.5 V		5.5 V	5.5 V		"		5.5 V	5.5 V		5.5 V	5.5 V	"	1A	2/	2/	"
		"	38	5.5 V	0.4 V		5.5 V	"		"		"	"		"	"	"	1B	"	"	"
		"	39	"	5.5 V		0.4 V	"		"		"	"		"	"	"	2A	"	"	"
		"	40	"	"		5.5 V	0.4 V		"		"	"		"	"	"	2B	"	"	"
		"	41	"	"		"	5.5 V		"		0.4 V	"		"	"	"	3A	"	"	"
		"	42	"	"		"	"		"		5.5 V	0.4 V		"	"	"	3B	"	"	"
		"	43	"	"		"	"		"		"	5.5 V		0.4 V	"	"	4A	"	"	"
		"	44	"	"		"	"		"		"	5.5 V		5.5 V	0.4 V	"	4B	"	"	"

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	1A	1B	1Y	2A	2B	2Y	GND	3Y	3A	3B	4Y	4A	4B	V_{CC}		Min	Max	
1 Tc = 25°C	I_O 3/	3011	45	GND	GND	2.25 V				GND							5.5 V	1Y	-15 4/	-70 4/	mA
		"	46				GND	GND	2.25 V	"							"	2Y	"	"	"
		"	47							"	2.25 V	GND	GND				"	3Y	"	"	"
		"	48							"				2.25 V	GND	GND	"	4Y	"	"	"
	I_{CCH}	3005	49	GND	GND		GND	GND		"		GND	GND		GND	GND	"	V_{CC}		0.85	"
	I_{CCL}	3005	50	4.5 V	4.5 V		4.5 V	4.5 V		"		4.5 V	4.5 V		4.5 V	4.5 V	"	V_{CC}		3.0	"
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9 Tc = 25°C	t_{PHL}	3003 Fig. 3	51	IN	2.7 V	OUT	2.7 V	2.7 V		GND		2.7 V	2.7 V		2.7 V	2.7 V	5.0 V	1A to 1Y	2	8	ns
			52	2.7 V	IN	OUT	2.7 V	"		"		"	"		"	"	"	1B to 1Y	"	"	"
		"	53	"	2.7 V		IN	"	OUT	"		"	"		"	"	"	2A to 2Y	"	"	"
		"	54	"	"		2.7 V	IN	OUT	"		"	"		"	"	"	2B to 2Y	"	"	"
		"	55	"	"		"	2.7 V		"	OUT	IN	"		"	"	"	3A to 3Y	"	"	"
		"	56	"	"		"	"		"	OUT	2.7 V	IN		"	"	"	3B to 3Y	"	"	"
		"	57	"	"		"	"		"		"	2.7 V	OUT	IN	"	"	4A to 4Y	"	"	"
		"	58	"	"		"	"		"		"	"	OUT	2.7 V	IN	"	4B to 4Y	"	"	"
	t_{PLH}	"	59	IN	"	OUT	"	"		GND		"	"		"	2.7 V	"	1A to 1Y	3	11	"
		"	60	2.7 V	IN	OUT	"	"		"		"	"		"	"	"	1B to 1Y	"	"	"
		"	61	"	2.7 V		IN	"	OUT	"		"	"		"	"	"	2A to 2Y	"	"	"
		"	62	"	"		2.7 V	IN	OUT	"		"	"		"	"	"	2B to 2Y	"	"	"
		"	63	"	"		"	2.7 V		"	OUT	IN	"		"	"	"	3A to 3Y	"	"	"
		"	64	"	"		"	"		"	OUT	2.7 V	IN		"	"	"	3B to 3Y	"	"	"
		"	65	"	"		"	"		"		"	2.7 V	OUT	IN	"	"	4A to 4Y	"	"	"
		"	66	"	"		"	"		"		"	"	OUT	2.7 V	IN	"	4B to 4Y	"	"	"
10	t_{PHL}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																	2	10	"
	t_{PLH}																		3	14	"
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

1/ For case 2, pins not referenced are N/C.

2/ I_{IL} limits shall be as follows:

Parameters	Min/Max limits in µA for circuit		
	A	B	C
I_{IL}	0/-100	0/-100	-0.2/-400

3/ Method 3011 of MIL-STD-883 shall be used, except the output shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current, I_{OS}.

4/ I_O limits for circuit A shall be -30 to -112 mA and for circuit C shall be -30 to -110 mA.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	1A	1B	2A	2B	2C	2Y	GND	3Y	3A	3B	3C	1Y	1C	V_{CC}		Min	Max	
1 Tc = 25°C	V_{OH}	3006	1	0.8 V	5.5 V	5.5 V	5.5 V	5.5 V		GND		5.5 V	5.5 V	5.5 V	-400 µA	5.5 V	4.5 V	1Y	2.5		V
		"	2	5.5 V	0.8 V	"	"	"		"		"	"	"	-400 µA	5.5 V	"	1Y	"		"
		"	3	"	5.5 V	"	"	"		"		"	"	"	-400 µA	0.8 V	"	1Y	"		"
		"	4	"	"	0.8 V	"	"	-400 µA	"		"	"	"		5.5 V	"	2Y	"		"
		"	5	"	"	5.5 V	0.8 V	"	-400 µA	"		"	"	"		"	"	2Y	"		"
		"	6	"	"	"	5.5 V	0.8 V	-400 µA	"		"	"	"		"	"	2Y	"		"
		"	7	"	"	"	"	5.5 V		"	-400 µA	0.8 V	"	"		"	"	3Y	"		"
		"	8	"	"	"	"	"		"	-400 µA	5.5 V	0.8 V	"		"	"	3Y	"		"
		"	9	"	"	"	"	"		"	-400 µA	5.5 V	5.5 V	0.8 V		"	"	3Y	"		"
	V_{OL}	3007	10	2.0 V	2.0 V	GND	GND	GND		"		GND	GND	GND	4 mA	2.0 V	"	1Y		0.4	"
		"	11	GND	GND	2.0 V	2.0 V	2.0 V	4 mA	"		GND	GND	GND		GND	"	2Y		"	"
		"	12	GND	GND	GND	GND	GND		"	4 mA	2.0 V	2.0 V	2.0 V		GND	"	3Y		"	"
	$V_{I C}$		13	-18 mA						"							"	1A		-1.5	"
			14		-18 mA					"							"	1B		"	"
			15			-18 mA				"							"	2A		"	"
			16				-18 mA			"							"	2B		"	"
			17					-18 mA		"							"	2C		"	"
			18							"		-18 mA					"	3A		"	"
			19							"			-18 mA				"	3B		"	"
			20							"				-18 mA			"	3C		"	"
			21							"						-18 mA	"	1C		"	"
	I_{IH1}	3010	22	2.7 V	GND	GND	GND	GND		"		GND	GND	GND		GND	5.5 V	1A		20	µA
		"	23	GND	2.7 V	GND	"	"		"		"	"	"		GND	"	1B		"	"
		"	24	"	GND	GND	"	"		"		"	"	"		2.7 V	"	1C		"	"
		"	25	"	"	2.7 V	"	"		"		"	"	"		GND	"	2A		"	"
		"	26	"	"	GND	2.7 V	"		"		"	"	"		"	"	2B		"	"
		"	27	"	"	"	GND	2.7 V		"		"	"	"		"	"	2C		"	"
		"	28	"	"	"	"	GND		"		2.7 V	"	"		"	"	3A		"	"
		"	29	"	"	"	"	"		"		GND	2.7 V	"		"	"	3B		"	"
		"	30	"	"	"	"	"		"		"	GND	2.7 V		"	"	3C		"	"
	I_{IH2}	"	31	7.0 V	"	"	"	"		"		"	"	GND		"	"	1A		100	"
		"	32	GND	7.0 V	"	"	"		"		"	"	"		"	"	1B		"	"
		"	33	"	GND	"	"	"		"		"	"	"		7.0 V	"	1C		"	"
		"	34	"	"	7.0 V	"	"		"		"	"	"		GND	"	2A		"	"
		"	35	"	"	GND	7.0 V	"		"		"	"	"		"	"	2B		"	"
		"	36	"	"	"	GND	7.0 V		"		"	"	"		"	"	2C		"	"
		"	37	"	"	"	"	GND		"		7.0 V	"	"		"	"	3A		"	"
		"	38	"	"	"	"	"		"		GND	7.0 V	"		"	"	3B		"	"
		"	39	"	"	"	"	"		"		GND	GND	7.0 V		"	"	3C		"	"
	I_{IL}	3009	40	0.4 V	5.5 V	5.5 V	5.5 V	5.5 V		"		5.5 V	5.5 V	5.5 V		5.5 V	"	1A	2/	2/	"
		"	41	5.5 V	0.4 V	"	"	"		"		"	"	"		5.5 V	"	1B	"	"	"
		"	42	"	5.5 V	"	"	"		"		"	"	"		0.4 V	"	1C	"	"	"
		"	43	"	"	0.4 V	"	"		"		"	"	"		5.5 V	"	2A	"	"	"
		"	44	"	"	5.5 V	0.4 V	"		"		"	"	"		"	"	2B	"	"	"
		"	45	"	"	"	5.5 V	0.4 V		"		"	"	"		"	"	2C	"	"	"
		"	46	"	"	"	"	5.5 V		"		0.4 V	"	"		"	"	3A	"	"	"
		"	47	"	"	"	"	"		"		5.5 V	0.4 V	"		"	"	3B	"	"	"
		"	48	"	"	"	"	"		"		"	5.5 V	0.4 V		"	"	3C	"	"	"

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	1A	1B	2A	2B	2C	2Y	GND	3Y	3A	3B	3C	1Y	1C	V_{CC}		Min	Max	
1 Tc = 25°C	I_O 3/	3011	49	GND	GND					GND					2.25 V	GND	5.5 V	1Y	-30 4/	-112 4/	mA
		"	50			GND	GND	GND	2.25 V	"							"	2Y	"	"	"
		"	51							"	2.25 V	GND	GND	GND			"	3Y	"	"	"
	I_{CCH}	3005	52	GND	GND	GND	GND	GND		"		GND	GND	GND		GND	"	V_{CC}		0.6	"
	I_{CCL}	3005	53	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V		"		4.5 V	4.5 V	4.5 V		4.5 V	"	V_{CC}		2.2	"
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9 Tc = 25°C	t_{PHL}	3003 Fig. 3	54	IN	2.7 V	2.7 V	2.7 V	2.7 V		GND		2.7 V	2.7 V	2.7 V	OUT	2.7 V	5.0 V	1A to 1Y	3	11	ns
			55	2.7 V	IN	2.7 V	"	"		"		"	"	"	OUT	2.7 V	"	1B to 1Y	"	"	"
		"	56	"	2.7 V	2.7 V	"	"		"		"	"	"	OUT	IN	"	1C to 1Y	"	"	"
		"	57	"	"	IN	"	"	OUT	"		"	"	"		2.7 V	"	2A to 2Y	"	"	"
		"	58	"	"	2.7 V	IN	"	OUT	"		"	"	"		"	"	2B to 2Y	"	"	"
		"	59	"	"	"	2.7 V	IN	OUT	"		"	"	"		"	"	2C to 2Y	"	"	"
		"	60	"	"	"	"	2.7 V		"	OUT	IN	"	"		"	"	3A to 3Y	"	"	"
		"	61	"	"	"	"	"		"	OUT	2.7 V	IN	"		"	"	3B to 3Y	"	"	"
		"	62	"	"	"	"	"		"	OUT	"	2.7 V	IN		"	"	3C to 3Y	"	"	"
	t_{PLH}	"	63	IN	"	"	"	"		"		"	"	2.7 V	OUT	"	"	1A to 1Y	3	12	"
		"	64	2.7 V	IN	"	"	"		"		"	"	"	OUT	"	"	1B to 1Y	"	"	"
		"	65	"	2.7 V	"	"	"		"		"	"	"	OUT	IN	"	1C to 1Y	"	"	"
		"	66	"	"	IN	"	"	OUT	"		"	"	"		2.7 V	"	2A to 2Y	"	"	"
		"	67	"	"	2.7 V	IN	"	OUT	"		"	"	"		"	"	2B to 2Y	"	"	"
		"	68	"	"	"	2.7 V	IN	OUT	"		"	"	"		"	"	2C to 2Y	"	"	"
		"	69	"	"	"	"	2.7 V		"	OUT	IN	"	"		"	"	3A to 3Y	"	"	"
		"	70	"	"	"	"	"		"	OUT	2.7 V	IN	"		"	"	3B to 3Y	"	"	"
		"	71	"	"	"	"	"		"	OUT	"	2.7 V	IN		"	"	3C to 3Y	"	"	"
10	t_{PHL}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																	3	12	"
	t_{PLH}																		3	15	"
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

1/ For case 2, pins not referenced are N/C.

2/ I_{IL} limits shall be as follows:

Parameters	Min/Max limits in µA for circuit		
	A	B	C
I_{IL}	0/-100	0/-100	-0.2/-400

3/ Method 3011 of MIL-STD-883 shall be used, except the output shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current, I_{OS}.

4/ I_O limits shall be -20/-112 mA for circuit A and -30 to -110 mA for circuit C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	1A	1B	NC	1C	1D	1Y	GND	2Y	2A	2B	NC	2C	2D	V_{CC}		Min	Max	
1 Tc = 25°C	V_{OH}	3006	1	0.8 V	5.5 V		5.5 V	5.5 V	-400 µA	GND		5.5 V	5.5 V		5.5 V	5.5 V	4.5 V	1Y	2.5		V
		"	2	5.5 V	0.8 V		"	"	"	"		"	"		"	"	"	1Y	"		"
		"	3	"	5.5 V		0.8 V	"	"	"		"	"		"	"	"	1Y	"		"
		"	4	"	"		5.5 V	0.8 V	"	"		"	"		"	"	"	1Y	"		"
		"	5	"	"		"	5.5 V		"	-400 µA	0.8 V	"		"	"	"	2Y	"		"
		"	6	"	"		"	"		"	"	5.5 V	0.8 V		"	"	"	2Y	"		"
		"	7	"	"		"	"		"	"	"	5.5 V		0.8 V	"	"	2Y	"		"
		"	8	"	"		"	"		"	"	"	5.5 V		5.5 V	0.8 V	"	2Y	"		"
	V_{OL}	3007	9	2.0 V	2.0 V		2.0 V	2.0 V	4 mA	"		GND	GND		GND	GND	"	1Y		0.4	"
		"	10	GND	GND		GND	GND		"	4 mA	2.0 V	2.0 V		2.0 V	2.0 V	"	2Y		0.4	"
	V_{IC}		11	-18 mA						"							"	1A		-1.5	"
			12		-18 mA					"							"	1B		"	"
			13				-18 mA			"							"	1C		"	"
			14					-18 mA		"							"	1D		"	"
			15							"		-18 mA					"	2A		"	"
			16							"			-18 mA				"	2B		"	"
			17							"					-18 mA		"	2C		"	"
			18							"						-18 mA	"	2D		"	"
	I_{IH1}	3010	19	2.7 V	GND		GND	GND		"		GND	GND		GND	GND	5.5 V	1A		20	µA
		"	20	GND	2.7 V		GND	"		"		"	"		"	"	"	1B		"	"
		"	21	"	GND		2.7 V	"		"		"	"		"	"	"	1C		"	"
		"	22	"	"		GND	2.7 V		"		"	"		"	"	"	1D		"	"
		"	23	"	"		"	GND		"		2.7 V	"		"	"	"	2A		"	"
		"	24	"	"		"	"		"		GND	2.7 V		"	"	"	2B		"	"
		"	25	"	"		"	"		"		"	GND		2.7 V	"	"	2C		"	"
		"	26	"	"		"	"		"		"	"		GND	2.7 V	"	2D		"	"
	I_{IH2}	"	27	7.0 V	"		"	"		"		"	"		"	GND	"	1A		100	"
		"	28	GND	7.0 V		"	"		"		"	"		"	"	"	1B		"	"
		"	29	"	GND		7.0 V	"		"		"	"		"	"	"	1C		"	"
		"	30	"	"		GND	7.0 V		"		"	"		"	"	"	1D		"	"
		"	31	"	"		"	GND		"		7.0 V	"		"	"	"	2A		"	"
		"	32	"	"		"	"		"		GND	7.0 V		"	"	"	2B		"	"
		"	33	"	"		"	"		"		"	GND		7.0 V	"	"	2C		"	"
		"	34	"	"		"	"		"		"	"		GND	7.0 V	"	2D		"	"
	I_{IL}	3009	35	0.4 V	5.5 V		5.5 V	5.5 V		"		5.5 V	5.5 V		5.5 V	5.5 V	"	1A	2/	2/	"
		"	36	5.5 V	0.4 V		5.5 V	"		"		"	"		"	"	"	1B	"	"	"
		"	37	"	5.5 V		0.4	"		"		"	"		"	"	"	1C	"	"	"
		"	38	"	"		5.5 V	0.4 V		"		"	"		"	"	"	1D	"	"	"
		"	39	"	"		"	5.5 V		"		0.4 V	"		"	"	"	2A	"	"	"
		"	40	"	"		"	"		"		5.5 V	0.4 V		"	"	"	2B	"	"	"
		"	41	"	"		"	"		"		"	5.5 V		0.4 V	"	"	2C	"	"	"
		"	42	"	"		"	"		"		"	5.5 V		5.5 V	0.4 V	"	2D	"	"	"
	I_O 3/	3011	43	GND	GND		GND	GND	2.25 V	"							"	1Y	-15 4/	-70 4/	mA
		"	44							"	2.25 V	GND	GND		GND	GND	"	2Y	-15 4/	-70 4/	mA

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 03 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	1A	1B	NC	1C	1D	1Y	GND	2Y	2A	2B	NC	2C	2D	V_{CC}		Min	Max	
1	I_{CCH}	3005	45	GND	GND		GND	GND		GND		GND	GND		GND	GND	5.5 V	V_{CC}		0.4	mA
Tc = 25°C	I_{CCL}	3005	46	4.5 V	4.5 V		4.5 V	4.5 V		GND		4.5 V	4.5 V		4.5 V	4.5 V	5.5 V	V_{CC}		1.5	mA
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9	t_{PHL}	3003	47	IN	2.7 V		2.7 V	2.7 V	OUT	GND		2.7 V	2.7 V		2.7 V	2.7 V	5.0 V	1A to 1Y	3	10	ns
Tc = 25°C		Fig. 3	48	2.7 V	IN		2.7 V	"	"	"		"	"		"	"	"	1B to 1Y	"	"	"
		"	49	"	2.7 V		IN	"	"	"		"	"		"	"	"	1C to 1Y	"	"	"
		"	50	"	"		2.7 V	IN	"	"		"	"		"	"	"	1D to 1Y	"	"	"
		"	51	"	"		"	2.7 V		"	OUT	IN	"		"	"	"	2A to 2Y	"	"	"
		"	52	"	"		"	"		"	"	2.7 V	IN		"	"	"	2B to 2Y	"	"	"
		"	53	"	"		"	"		"	"	"	2.7 V		IN	"	"	2C to 2Y	"	"	"
		"	54	"	"		"	"		"	"	"	"		2.7 V	IN	"	2D to 2Y	"	"	"
	t_{PLH}	"	55	IN	"		"	"	OUT	"		"	"		"	2.7 V	"	1A to 1Y	3	11	"
		"	56	2.7 V	IN		"	"	"	"		"	"		"	"	"	1B to 1Y	"	"	"
		"	57	"	2.7 V		IN	"	"	"		"	"		"	"	"	1C to 1Y	"	"	"
		"	58	"	"		2.7 V	IN	"	"		"	"		"	"	"	1D to 1Y	"	"	"
		"	59	"	"		"	2.7 V		"	OUT	IN	"		"	"	"	2A to 2Y	"	"	"
		"	60	"	"		"	"		"	"	2.7 V	IN		"	"	"	2B to 2Y	"	"	"
		"	61	"	"		"	"		"	"	"	2.7 V		IN	"	"	2C to 2Y	"	"	"
		"	62	"	"		"	"		"	"	"	"		2.7 V	IN	"	2D to 2Y	"	"	"
10	t_{PHL}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																	3	15	"
	t_{PLH}																		3	13	"
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

1/ For case 2, pins not referenced are N/C.

2/ I_{IL} limits shall be as follows:

Parameters	Min/Max limits in µA for circuit		
	A	B	C
I_{IL}	0/-100	0/-100	-0.2/-400

3/ Method 3011 of MIL-STD-883 shall be used, except the output shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current, I_{OS}.

4/ I_O limits shall be -20 to -112 mA for circuit A and -30 to -110 mA and for circuit C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	A	B	C	D	E	F	GND	Y	NC	NC	G	H	NC	V_{CC}		Min	Max	
1 Tc = 25°C	V_{OH}	3006	1	0.8 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND	-400 µA			5.5 V	5.5 V		4.5 V	Y	2.5		V
		"	2	5.5 V	0.8 V	5.5 V	"	"	"	"	"			"	"		"	Y	"		"
		"	3	"	5.5 V	0.8 V	"	"	"	"	"			"	"		"	Y	"		"
		"	4	"	"	5.5 V	0.8 V	"	"	"	"			"	"		"	Y	"		"
		"	5	"	"	"	5.5 V	0.8 V	"	"	"			"	"		"	Y	"		"
		"	6	"	"	"	"	5.5 V	0.8 V	"	"			"	"		"	Y	"		"
		"	7	"	"	"	"	"	5.5 V	"	"			0.8 V	"		"	Y	"		"
		"	8	"	"	"	"	"	5.5 V	"	"			5.5 V	0.8 V		"	Y	"		"
	V_{OL}	3007	9	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	"	4 mA			2.0 V	2.0 V		"	Y		0.4	"
	V_{IC}		10	-18 mA						"							"	A		-1.5	"
			11		-18 mA					"							"	B		"	"
			12			-18 mA				"							"	C		"	"
			13				-18 mA			"							"	D		"	"
			14					-18 mA		"							"	E		"	"
			15						-18 mA	"							"	F		"	"
			16							"				-18 mA			"	G		"	"
			17							"					-18 mA		"	H		"	"
	I_{IH1}	3010	18	2.7 V	GND	GND	GND	GND	GND	"				GND	GND		5.5 V	A		20	µA
		"	19	GND	2.7 V	GND	"	"	"	"				"	"		"	B		"	"
		"	20	"	GND	2.7 V	"	"	"	"				"	"		"	C		"	"
		"	21	"	"	GND	2.7 V	"	"	"				"	"		"	D		"	"
		"	22	"	"	"	GND	2.7 V	"	"				"	"		"	E		"	"
		"	23	"	"	"	"	GND	2.7 V	"				"	"		"	F		"	"
		"	24	"	"	"	"	"	GND	"				2.7 V	"		"	G		"	"
		"	25	"	"	"	"	"	"	"				GND	2.7 V		"	H		"	"
	I_{IH2}	"	26	7.0 V	"	"	"	"	"	"				"	GND		"	A		100	"
		"	27	GND	7.0 V	"	"	"	"	"				"	"		"	B		"	"
		"	28	"	GND	7.0 V	"	"	"	"				"	"		"	C		"	"
		"	29	"	"	GND	7.0 V	"	"	"				"	"		"	D		"	"
		"	30	"	"	"	GND	7.0 V	"	"				"	"		"	E		"	"
		"	31	"	"	"	"	GND	7.0 V	"				"	"		"	F		"	"
		"	32	"	"	"	"	"	GND	"				7.0 V	"		"	G		"	"
		"	33	"	"	"	"	"	GND	"				GND	7.0 V		"	H		"	"
	I_{IL}	3009	34	0.4 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"				5.5 V	5.5 V		"	A	2/	2/	"
			35	5.5 V	0.4 V	5.5 V	"	"	"	"				"	"		"	B	"	"	"
		"	36	"	5.5 V	0.4 V	"	"	"	"				"	"		"	C	"	"	"
		"	37	"	"	5.5 V	0.4 V	"	"	"				"	"		"	D	"	"	"
		"	38	"	"	"	5.5 V	0.4 V	"	"				"	"		"	E	"	"	"
		"	39	"	"	"	"	5.5 V	0.4 V	"				"	"		"	F	"	"	"
		"	40	"	"	"	"	"	5.5 V	"				0.4 V	"		"	G	"	"	"
		"	41	"	"	"	"	"	5.5 V	"				5.5 V	0.4 V		"	H	"	"	"
	I_O 3/	3011	42	GND	GND	GND	GND	GND	GND	"	2.25 V			GND	GND		"	Y	-30 4/	-112 4/	mA
	I_{CCH}	3005	43	GND	GND	GND	GND	GND	GND	"				GND	GND		"	V_{CC}		360	µA
	I_{CCL}	3005	44	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"				4.5 V	4.5 V		"	V_{CC}		0.9	mA

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 04 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	A	B	C	D	E	F	GND	Y	NC	NC	G	H	NC	V_{CC}		Min	Max	
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9	t_{PHL}	3003	45	IN	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	GND	OUT			2.7 V	2.7 V		5.0 V	A to Y	5	20	ns
Tc = 25°C		Fig. 3	46	2.7 V	IN	2.7 V	"	"	"	"	"			"	"		"	B to Y	"	"	"
		"	47	"	2.7 V	IN	"	"	"	"	"			"	"		"	C to Y	"	"	"
		"	48	"	"	2.7 V	IN	"	"	"	"			"	"		"	D to Y	"	"	"
		"	49	"	"	"	2.7 V	IN	"	"	"			"	"		"	E to Y	"	"	"
		"	50	"	"	"	"	2.7 V	IN	"	"			"	"		"	F to Y	"	"	"
		"	51	"	"	"	"	"	2.7 V	"	"			IN	"		"	G to Y	"	"	"
		"	52	"	"	"	"	"	"	"	"			2.7 V	IN		"	H to Y	"	"	"
	t_{PLH}	"	53	IN	"	"	"	"	"	"	"			"	2.7 V		"	A to Y	3	10	"
		"	54	2.7 V	IN	"	"	"	"	"	"			"	"		"	B to Y	"	"	"
		"	55	"	2.7 V	IN	"	"	"	"	"			"	"		"	C to Y	"	"	"
		"	56	"	"	2.7 V	IN	"	"	"	"			"	"		"	D to Y	"	"	"
		"	57	"	"	"	2.7 V	IN	"	"	"			"	"		"	E to Y	"	"	"
		"	58	"	"	"	"	2.7 V	IN	"	"			"	"		"	F to Y	"	"	"
		"	59	"	"	"	"	"	2.7 V	"	"			IN	"		"	G to Y	"	"	"
		"	60	"	"	"	"	"	"	"	"			2.7 V	IN		"	H to Y	"	"	"
10	t_{PHL}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																	5	22	"
	t_{PLH}																		3	12	"
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

1/ For case 2, pins not referenced are N/C.

2/ I_{IL} limits shall be as follows:

Parameters	Min/Max limits in µA for circuit		
	A	B	C
I_{IL}	0/-100	0/-100	-0.2/-400

3/ Method 3011 of MIL-STD-883 shall be used, except the output shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current, I_{OS}.

4/ I_O limits shall be -20 to -112 mA for circuit A and -30 to -110 mA and for circuit C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 05.

Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 2 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	A	B	C	D	E	F	G	GND	Y	H	I	J	K	L	M	V_{CC}		Min	Max	
1 Tc = 25°C	V_{OH}	3006	1	0.8 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND	-400 µA	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	4.5 V	Y	2.5		V
		"	2	5.5 V	0.8 V	5.5 V	"	"	"	"	"	"	"	"	"	"	"	"	"	Y	"		"
		"	3	"	5.5 V	0.8 V	"	"	"	"	"	"	"	"	"	"	"	"	"	Y	"		"
		"	4	"	"	5.5 V	0.8 V	"	"	"	"	"	"	"	"	"	"	"	"	Y	"		"
		"	5	"	"	"	5.5 V	0.8 V	"	"	"	"	"	"	"	"	"	"	"	Y	"		"
		"	6	"	"	"	"	5.5 V	0.8 V	"	"	"	"	"	"	"	"	"	"	Y	"		"
		"	7	"	"	"	"	"	5.5 V	0.8 V	"	"	"	"	"	"	"	"	"	Y	"		"
		"	8	"	"	"	"	"	"	5.5 V	"	"	0.8 V	"	"	"	"	"	"	Y	"		"
		"	9	"	"	"	"	"	"	"	"	"	5.5 V	0.8 V	"	"	"	"	"	Y	"		"
		"	10	"	"	"	"	"	"	"	"	"	"	5.5 V	0.8 V	"	"	"	"	Y	"		"
		"	11	"	"	"	"	"	"	"	"	"	"	"	5.5 V	0.8 V	"	"	"	Y	"		"
		"	12	"	"	"	"	"	"	"	"	"	"	"	"	5.5 V	0.8 V	"	"	Y	"		"
		"	13	"	"	"	"	"	"	"	"	"	"	"	"	5.5 V	5.5 V	0.8 V	"	Y	"		"
	V_{OL}	3007	14	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	"	4 mA	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	"	Y		0.4	"
	V_{IC}		15	-18 mA							"								"	A		-1.5	"
			16		-18 mA						"								"	B		"	"
			17			-18 mA					"								"	C		"	"
			18				-18 mA				"								"	D		"	"
			19					-18 mA			"								"	E		"	"
			20						-18 mA		"								"	F		"	"
			21							-18 mA	"								"	G		"	"
			22								"		-18 mA						"	H		"	"
			23								"			-18 mA					"	I		"	"
			24								"				-18 mA				"	J		"	"
			25								"					-18 mA			"	K		"	"
			26								"						-18 mA		"	L		"	"
			27								"							-18 mA	"	M		"	"
	I_{IH1}	3010	28	2.7 V	GND	GND	GND	GND	GND	GND	"		GND	GND	GND	GND	GND	GND	5.5 V	A		20	µA
		"	29	GND	2.7 V	GND	"	"	"	"	"		"	"	"	"	"	"	"	B		"	"
		"	30	"	GND	2.7 V	"	"	"	"	"		"	"	"	"	"	"	"	C		"	"
		"	31	"	"	GND	2.7 V	"	"	"	"		"	"	"	"	"	"	"	D		"	"
		"	32	"	"	"	GND	2.7 V	"	"	"		"	"	"	"	"	"	"	E		"	"
		"	33	"	"	"	"	GND	2.7 V	"	"		"	"	"	"	"	"	"	F		"	"
		"	34	"	"	"	"	"	GND	2.7 V	"		"	"	"	"	"	"	"	G		"	"
		"	35	"	"	"	"	"	"	GND	"		2.7 V	"	"	"	"	"	"	H		"	"
		"	36	"	"	"	"	"	"	"	"		GND	2.7 V	"	"	"	"	"	I		"	"
		"	37	"	"	"	"	"	"	"	"		"	GND	2.7 V	"	"	"	"	J		"	"
		"	38	"	"	"	"	"	"	"	"		"	"	GND	2.7 V	"	"	"	K		"	"
		"	39	"	"	"	"	"	"	"	"		"	"	"	GND	2.7 V	"	"	L		"	"
		"	40	"	"	"	"	"	"	"	"		"	"	"	"	GND	2.7 V	"	M		"	"
	I_{IH2}	"	41	7.0 V	"	"	"	"	"	"	"		"	"	"	"	"	GND	"	A		100	"
		"	42	GND	7.0 V	"	"	"	"	"	"		"	"	"	"	"	"	"	B		"	"
		"	43	"	GND	7.0 V	"	"	"	"	"		"	"	"	"	"	"	"	C		"	"
		"	44	"	"	GND	7.0 V	"	"	"	"		"	"	"	"	"	"	"	D		"	"
		"	45	"	"	"	GND	7.0 V	"	"	"		"	"	"	"	"	"	"	E		"	"
		"	46	"	"	"	"	GND	7.0 V	"	"		"	"	"	"	"	"	"	F		"	"
		"	47	"	"	"	"	"	GND	7.0 V	"		"	"	"	"	"	"	"	G		"	"
		"	48	"	"	"	"	"	"	GND	"		7.0 V	"	"	"	"	"	"	H		"	"
		"	49	"	"	"	"	"	"	"	"		GND	7.0 V	"	"	"	"	"	I		"	"
		"	50	"	"	"	"	"	"	"	"		"	GND	7.0 V	"	"	"	"	J		"	"
		"	51	"	"	"	"	"	"	"	"		"	"	GND	7.0 V	"	"	"	K		"	"
		"	52	"	"	"	"	"	"	"	"		"	"	"	GND	7.0 V	"	"	L		"	"
		"	53	"	"	"	"	"	"	"	"		"	"	"	GND	GND	7.0 V	"	M		"	"

See footnotes at end of table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 2 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	A	B	C	D	E	F	G	GND	Y	H	I	J	K	L	M	V_{CC}		Min	Max	
1	I_{IL}	3009	54	0.4 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND		5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	A	2/	2/	µA
Tc = 25°C		"	55	5.5 V	0.4 V	5.5 V	"	"	"	"	"		"	"	"	"	"	"	"	B	"	"	"
		"	56	"	5.5 V	0.4 V	"	"	"	"	"		"	"	"	"	"	"	"	C	"	"	"
		"	57	"	"	5.5 V	0.4 V	"	"	"	"		"	"	"	"	"	"	"	D	"	"	"
		"	58	"	"	"	5.5 V	0.4 V	"	"	"		"	"	"	"	"	"	"	E	"	"	"
		"	59	"	"	"	"	5.5 V	0.4 V	"	"		"	"	"	"	"	"	"	F	"	"	"
		"	60	"	"	"	"	"	5.5 V	0.4 V	"		"	"	"	"	"	"	"	G	"	"	"
		"	61	"	"	"	"	"	"	5.5 V	"		0.4 V	"	"	"	"	"	"	H	"	"	"
		"	62	"	"	"	"	"	"	"	"		5.5 V	0.4 V	"	"	"	"	"	I	"	"	"
		"	63	"	"	"	"	"	"	"	"		"	5.5 V	0.4 V	"	"	"	"	J	"	"	"
		"	64	"	"	"	"	"	"	"	"		"	"	5.5 V	0.4 V	"	"	"	K	"	"	"
		"	65	"	"	"	"	"	"	"	"		"	"	"	5.5 V	0.4 V	"	"	L	"	"	"
		"	66	"	"	"	"	"	"	"	"		"	"	"	"	5.5 V	0.4 V	"	M	"	"	"
	I_O 3/	3011	67	GND	GND	GND	GND	GND	GND	GND	"	2.25 V	GND	GND	GND	GND	GND	GND	"	Y	-30 4/	-112 4/	mA
	I_{CCH}	3005	68	GND	GND	GND	GND	GND	GND	GND	"		GND	GND	GND	GND	GND	GND	"	V_{CC}		340	µA
	I_{CCL}	3005	69	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"		4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	V_{CC}		800	µA
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
9	t_{PHL}	3003	70	IN	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	GND	OUT	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	5.0 V	A to Y	5	25	ns
T_C = 25°C		Fig. 3	71	2.7 V	IN	2.7 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B to Y	"	"	"
		"	72	"	2.7 V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	"	C to Y	"	"	"
		"	73	"	"	2.7 V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	D to Y	"	"	"
		"	74	"	"	"	2.7 V	IN	"	"	"	"	"	"	"	"	"	"	"	"	E to Y	"	"	"
		"	75	"	"	"	"	2.7 V	IN	"	"	"	"	"	"	"	"	"	"	"	F to Y	"	"	"
		"	76	"	"	"	"	"	2.7 V	IN	"	"	"	"	"	"	"	"	"	"	G to Y	"	"	"
		"	77	"	"	"	"	"	"	2.7 V	"	"	IN	"	"	"	"	"	"	"	H to Y	"	"	"
		"	78	"	"	"	"	"	"	"	"	"	2.7 V	IN	"	"	"	"	"	"	I to Y	"	"	"
		"	79	"	"	"	"	"	"	"	"	"	"	2.7 V	IN	"	"	"	"	"	J to Y	"	"	"
		"	80	"	"	"	"	"	"	"	"	"	"	"	2.7 V	IN	"	"	"	"	K to Y	"	"	"
		"	81	"	"	"	"	"	"	"	"	"	"	"	"	2.7 V	IN	"	"	"	L to Y	"	"	"
		"	82	"	"	"	"	"	"	"	"	"	"	"	"	"	2.7 V	IN	"	"	M to Y	"	"	"
	t_{PLH}	"	83	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	2.7 V	"	"	A to Y	3	11	"
		"	84	2.7 V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B to Y	"	"	"
		"	85	"	2.7 V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	"	C to Y	"	"	"
		"	86	"	"	2.7 V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	D to Y	"	"	"
		"	87	"	"	"	2.7 V	IN	"	"	"	"	"	"	"	"	"	"	"	"	E to Y	"	"	"
		"	88	"	"	"	"	2.7 V	IN	"	"	"	"	"	"	"	"	"	"	"	F to Y	"	"	"
		"	89	"	"	"	"	"	2.7 V	IN	"	"	"	"	"	"	"	"	"	"	G to Y	"	"	"
		"	90	"	"	"	"	"	"	2.7 V	"	"	IN	"	"	"	"	"	"	"	H to Y	"	"	"
		"	91	"	"	"	"	"	"	"	"	"	2.7 V	IN	"	"	"	"	"	"	I to Y	"	"	"
		"	92	"	"	"	"	"	"	"	"	"	"	2.7 V	IN	"	"	"	"	"	J to Y	"	"	"
		"	93	"	"	"	"	"	"	"	"	"	"	"	2.7 V	IN	"	"	"	"	K to Y	"	"	"
		"	94	"	"	"	"	"	"	"	"	"	"	"	"	2.7 V	IN	"	"	"	L to Y	"	"	"
		"	95	"	"	"	"	"	"	"	"	"	"	"	"	"	2.7 V	IN	"	"	M to Y	"	"	"
10	t_{PHL}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.																				5	28	"
	t_{PLH}																					3	14	"
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																						

1/ For case 2, pins not referenced are N/C.

2/ I_{IL} limits shall be as follows:

Parameters	Min/Max limits in µA for circuit		
	A	B	C
I_{IL}	0/-100	0/-100	-0.2/-400

3/ Method 3011 of MIL-STD-883 shall be used, except the output shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current, I_{OS}.

4/ I_O limits shall be -20 to -112 mA for circuit A and -30 to -110 mA and for circuit C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 06.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	V_{CC}		Min	Max	
1 Tc = 25°C	V_{OH}	3006	1	0.8 V	-400 µA	5.5 V		5.5 V		GND		5.5 V		5.5 V		5.5 V	4.5 V	1Y	2.5		V
		"	2	5.5 V		0.8 V	-400 µA	5.5 V		"		"		"		"	"	2Y	"		"
		"	3	"		5.5 V		0.8 V	-400 µA	"		"		"		"	"	3Y	"		"
		"	4	"		"		5.5 V		"	-400 µA	0.8 V		"		"	"	4Y	"		"
		"	5	"		"		"		"		5.5 V	-400 µA	0.8 V		"	"	5Y	"		"
		"	6	"		"		"		"		5.5 V		5.5 V	-400 µA	0.8 V	"	6Y	"		"
	V_{OL}	3007	7	2.0 V	4 mA	GND		GND		"		GND		GND		GND	"	1Y		0.4	"
		"	8	GND		2.0 V	4 mA	GND		"		"		"		"	"	2Y		"	"
		"	9	"		GND		2.0 V	4 mA	"		"		"		"	"	3Y		"	"
		"	10	"		"		GND		"	4 mA	2.0 V		"		"	"	4Y		"	"
		"	11	"		"		"		"		GND	4 mA	2.0 V		"	"	5Y		"	"
		"	12	"		"		"		"		GND		GND	4 mA	2.0 V	"	6Y		"	"
	$V_{I C}$		13	-18 mA						"							"	1A		-1.5	"
			14			-18 mA				"							"	2A		"	"
			15					-18 mA		"							"	3A		"	"
			16							"		-18 mA					"	4A		"	"
			17							"				-18 mA			"	5A		"	"
			18							"						-18 mA	"	6A		"	"
	I_{IH1}	3010	19	2.7 V		GND		GND		"		GND		GND		GND	5.5 V	1A		20	µA
		"	20	GND		2.7 V		GND		"		"		"		"	"	2A		"	"
		"	21	"		GND		2.7 V		"		"		"		"	"	3A		"	"
		"	22	"		"		GND		"		2.7 V		"		"	"	4A		"	"
		"	23	"		"		"		"		GND		2.7 V		'	"	5A		"	"
		"	24	"		"		"		"		"		GND		2.7 V	"	6A		"	"
	I_{IH2}	"	25	7.0 V		"		"		"		"		"		GND	"	1A		100	"
		"	26	GND		7.0 V		"		"		"		"		"	"	2A		"	"
		"	27	"		GND		7.0 V		"		"		"		"	"	3A		"	"
		"	28	"		"		GND		"		7.0 V		"		"	"	4A		"	"
		"	29	"		"		"		"		GND		7.0 V		"	"	5A		"	"
		"	30	"		"		"		"		GND		GND		7.0 V	"	6A		"	"
	I_{IL}	3009	31	0.4 V		5.5 V		5.5 V		"		5.5 V		5.5 V		5.5 V	"	1A	2/	2/	"
		"	32	5.5 V		0.4 V		5.5 V		"		"		"		"	"	2A		"	"
		"	33	"		5.5 V		0.4 V		"		"		"		"	"	3A		"	"
		"	34	"		"		5.5 V		"		0.4 V		"		"	"	4A		"	"
		"	35	"		"		"		"		5.5 V		0.4 V		"	"	5A		"	"
		"	36	"		"		"		"		5.5 V		5.5 V		0.4 V	"	6A		"	"
	I_O 3/	3011	37	GND	2.25 V					"							"	1Y	-15 4/	-70 4/	mA
		"	38			GND	2.25 V			"							"	2Y	"	"	"
		"	39					GND	2.25 V	"							"	3Y	"	"	"
		"	40							"	2.25 V	GND					"	4Y	"	"	"
		"	41							"			2.25 V	GND			"	5Y	"	"	"
		"	42							"					2.25 V	GND	"	6Y	"	"	"
	I_{CCH}	3005	43	GND		GND		GND		"		GND		GND		GND	"	V_{CC}		1.1	"
	I_{CCL}	3005	44	4.5 V		4.5 V		4.5 V		"		4.5 V		4.5 V		4.5 V	"	V_{CC}		4.2	"

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	V_{CC}		Min	Max	
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9 Tc = 25°C	t_{PHL}	3003 Fig. 3	45	IN	OUT	2.7 V		2.7 V		GND		2.7 V		2.7 V		2.7 V	5.0 V	1A to 1Y	2	9	ns
		"	46	2.7 V		IN	OUT	2.7 V		"		"		"		"	"	2A to 2Y	"	"	"
		"	47	"		2.7 V		IN	OUT	"		"		"		"	"	3A to 3Y	"	"	"
		"	48	"		"		2.7 V		"	OUT	IN		"		"	"	4A to 4Y	"	"	"
		"	49	"		"		"		"		2.7 V	OUT	IN		"	"	5A to 5Y	"	"	"
		"	50	"		"		"		"		"		2.7 V	OUT	IN	"	6A to 6Y	"	"	"
	t_{PLH}	"	51	IN	OUT	"		"		"		"		"		2.7 V	"	1A to 1Y	3	11	"
		"	52	2.7 V		IN	OUT	"		"		"		"		"	"	2A to 2Y	"	"	"
		"	53	"		2.7 V		IN	OUT	"		"		"		"	"	3A to 3Y	"	"	"
		"	54	"		"		2.7 V		"	OUT	IN		"		"	"	4A to 4Y	"	"	"
		"	55	"		"		"		"		2.7 V	OUT	IN		"	"	5A to 5Y	"	"	"
		"	56	"		"		"		"		"		2.7 V	OUT	IN	"	6A to 6Y	"	"	"
10	t_{PHL}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																	2	12	"
	t_{PLH}																		3	17	"
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

1/ For case 2, pins not referenced are N/C.

2/ I_{IL} limits shall be as follows:

Parameters	Min/Max limits in µA for circuit		
	A	B	C
I_{IL}	0/-100	0/-100	-0.2/-400

3/ Method 3011 of MIL-STD-883 shall be used, except the output shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current, I_{OS}.

4/ I_O limits shall be -30 to -112 mA for circuit A and -30 to -110 mA and for circuit C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. PIN and compliance identifier, if applicable (see 1.2).

c. Requirements for delivery of one copy of the conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

d. Requirements for certificate of compliance, if applicable.

e. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

g. Requirements for product assurance options.

h. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements should not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

i. Requirements for "JAN" marking.

j. Packaging requirements (see 5.1).

6.3 Superseding information. The requirements of MIL-M-38510 have been superseded to take advantage of the available Qualified Manufacturer Listing (QML) system provided by MIL-PRF-38535. Previous references to MIL-M-38510 in this document have been replaced by appropriate references to MIL-PRF-38535. All technical requirements now consist of this specification and MIL-PRF-38535. The MIL-M-38510 specification sheet number and PIN have been retained to avoid adversely impacting existing government logistics systems and contractor's parts lists.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-38535 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DSCC-VQ, 3990 E. Broad Street, Columbus, Ohio 43123-1199.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

6.5 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-38535, MIL-HDBK-1331, and as follows:

GND ..... Ground zero voltage potential
V_{IN} ..... Voltage level at an input terminal
V_{IC} ..... Input clamp voltage
I_{IN} ..... Current flowing into an input terminal

6.6 Logistic support. Lead materials and finishes (see 3.4) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish A (see 3.4). Longer length leads and lead forming should not affect the part number.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-35810 device types and may have slight physical variations in relation to case size. The presence of this information should not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-PRF-38535.

Military device type	Generic-industry type
01	54ALS00A
02	54ALS10
03	54ALS20A
04	54ALS30
05	54ALS133
06	54ALS04A

6.8 Manufacturers' designation. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuits		
	A	B	C
	Texas Instruments	Motorola Inc.	National Semiconductor/ Fairchild Semiconductor
01	X	X	X
02	X	X	X
03	X	X	X
04	X	X	X
05	X		X
06	X	X	X

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5962-2053)

Review activities:
Army - MI, SM
Navy - AS, CG, MC, SH, TD
Air Force - 03, 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at www.dodssp.daps.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:28Z
Check the source to verify that this is the current version before use.